

KerrNullGeo

`KerrNullGeo` [*a*, *xs*, *ps*, *M_* : 1, *Options*]

returns a `KerrNullGeoFunction` which stores information about the trajectory of a light-ray starting from specified initial conditions. The black hole spin *a*, position *xs*, and wavevector *ps* are assumed to be given in units of the BH mass, unless mass *M* is specified (optional argument).

The following options can be given:

"Momentum"	"Momentum"	Specifies whether the user provided 4 - momentum or wave 4 - vector as <i>ps</i> . The default option is "Momentum" -> "Momentum". If the user provided the wave 4 - vector, the option "Momentum" -> "WaveVector" should be specified.
"PhiRange"	{-Infinity, Infinity}	This sets the range of output of the azimuthal angle. The default is "PhiRange" -> {-∞, ∞}, which starts the coordinate at 0 and does not take the modulus of it after full windings. Typical options could be {-π, π} or {0, 2 π}, but other option values in the format {bottomvalue, topvalue} are valid as well.

Tech Notes ⓘ
KerrNullGeodesics

Related Links ⓘ
XXXX

See Also ⓘ
KerrNullGeoDistant ▪ **KerrNullGeoFunction** ▪ ⓘ

Related Guides
KerrNullGeodesics

Examples Initialization ⓘ

Needs ["BlackHoleImages`"]

Basic Examples

[More Examples ▸](#)

Compute a geodesic in geometry given by $a = 0.9$, with some initial position and momentum:

```
In[13]:= geod = KerrNullGeo[0.9, {0, 150,  $\pi/3$ , 0}, {-1, -1, Sqrt[10], 10}];
```

Access the constant of motion l and the escape coordinates θ_x, ϕ_x :

```
In[14]:= l = geod["ConstantsOfMotion"] ["l"]  
          { $\theta_x$ ,  $\phi_x$ } = geod["EscapeCoordinates"]
```

```
Out[14]= 10
```

```
Out[15]= {1.99164, 3.49568}
```

Get the Boyer-Lindquist coordinates at Mino time $\lambda=0.1$:

```
In[16]:= geod[0.1]
```

```
Out[16]= {151.55, 12.0111, 1.60014, 1.12364}
```

More Examples ⓘ

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